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CGIAR Initiative on Fruit and Vegetables for Sustainable Healthy Diets

Piloting reusable plastic crates in the tomato value chain in Benin

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The FRESH Initiative activities are bundled into six packages, namely:

- Work Package 1: Understanding and Influencing Consumer Behaviour
- Work Package 2: Biodiversity, genetic innovation, and seed systems
- Work Package 3: Safe and sustainable production systems
- Work Package 4: Post-harvest and inclusive markets
- Work Package 5: Food Environments
- Work Package 6: Strengthening the enabling environment.

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Piloting reusable plastic crates in the tomato value chain in Benin

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Summary

Do reusable plastic crates perform better compared to traditional baskets?

The study highlights the challenges of introducing reusable plastic crates as an alternative packaging solution in South Benin's tomato value chain. The study produced mixed outcomes, making it difficult to state that reusable plastic crates perform better compared to traditional baskets.

Reusable plastic crates show potential for reducing post-harvest losses and increasing value in the tomato value chain, but their adoption faces challenges in cost and logistics, necessitating innovative business models for scalability.

The results showed variability in post-harvest losses across packaging methods. While reusable plastic crates demonstrated slightly lower post-harvest losses compared to traditional baskets, the differences were not substantial. Loss percentages from farm to the retail level were inconsistent, with in some cases no clear trend favoring crates over traditional methods.

Our study indicates a small opportunity to increase value across all stakeholders in the value chain by adopting improved packaging solutions. In the first round of assessments, four value chains demonstrated significantly higher monetary losses when using traditional baskets compared to reusable plastic crates. While crates preserved quality slightly better, the marginal improvement in prices did not offset the additional costs associated with the investment. Participants noted that the initial investment in crates posed a significant barrier, especially given the limited observed advantages during the pilot.

Feedback highlighted logistical challenges with crate availability and the need for better integration into existing workflows. Some participants had concerns about limited tomato availability during the study period and this further complicated the data collection, particularly during the second round of data collection.

Developing a sustainable business model for the returnable plastic crate intervention poses challenges. Implementing requires substantial investments, which are beyond the financial capacity of the participants. As a result, exploring alternative funding mechanisms or business models is essential to ensure the project's feasibility and long-term success.

Methodology

The study was conducted in Ouidah in Benin. Farmers in Ouidah receive support from the FRESH initiative. Ouidah has a large agricultural center where many different fruits and vegetables are produced and traded in large volumes. study also revealed limitations of the intervention due to contextual factors. The distance to the market might have influenced this and the proximity of farms to markets reduced the potential impact of improved packaging, as shorter transport durations minimized spoilage risks. The limited sample size, the differences in packing, the difference in weight between the units and seasonal constraints also limit the robustness of the findings.

Keywords:
Fruit and vegetables, post-harvest losses, value chain, tomato

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1. Introduction

Good postharvest management of vegetables is a critical challenge for value chain actors, as these crops exhibit significantly higher post-harvest losses compared to cereals or pulses. In Benin, tomatoes, characterized by a high-water content (85%-95%) (Aisso, et al., 2018), are particularly prone to spoilage due to inadequate packaging and poor handling during transportation and storage (Fassinou Hotegni et al. 2022). Traditional methods such as using traditional baskets lead to significant losses, estimated at 6.9% to 16.5% of the total volume from farm to wholesale levels (Kablan and Bancal, 2019). Additionally, damaged tomatoes are sold at reduced prices, further exacerbating economic losses for farmers and retailers. As a result, the post-harvest losses also contribute to economic losses for the farmers and the wholesalers. Using other packaging units such as reusable plastic crates for transportation and storage could offer a good alternative to reduce the post-harvest losses of tomatoes (Kok et al., 2019).

The adoption of innovative packaging solutions in Nigeria has significantly reduced post-harvest losses and improved product quality for perishable goods, including tomatoes (Plaisier et al., 2019; Kok et al 2019). These materials have proven particularly effective in maintaining the quality of tomatoes during transportation and storage, enhancing their market value. Given the similarities in agricultural practices and market conditions, introducing such packing materials in Benin presents a viable opportunity to address storage and transportation challenges.

This study assesses the effectiveness of reusable plastic crates as an alternative packaging solution to reduce post-harvest losses and increase economic value in South Benin's tomato value chain. In this report we will answer the following research question:

- Do reusable plastic crates perform better compared to traditional baskets?

Sub-research questions:

- Does the use of reusable plastic crates reduce tomato post-harvest losses compared to the use of traditional baskets?
- Does the use of reusable plastic crates increase the value of the tomatoes traded in the value chain compared to traditional baskets?
- How are the plastic crates perceived by the stakeholders in the tomato value chain and what are the opportunities for scaling?
- What is a feasible business model for the implementation of this innovation?

Piloting this approach strengths the value chain by reducing losses, improving the value in the entire value chain of fresh produce, and aligning with the broader goals of the CGIAR initiative on Fruit and Vegetables for Sustainable Healthy Diets (FRESH).

2. Methodology

2.1 Study area

The study was conducted in Ouidah in Benin. Farmers in Ouidah receive support from the FRESH initiative. Ouidah has a large agricultural center where many different fruits and vegetables are produced and traded in large volumes. The FRESH hub is accessible by a main road that has recently been built. The main fruit and vegetable crops cultivated are tomato, gboma, chilli, watermelon, vernonia, African basilic (tchiayo), amaranth, cucumber, cabbage, and onion (Egah et al., 2024). This study describes the trade dynamics in the area and explains, amongst others, that the unit of measurement of these products varies. Tomatoes are sold by stakeholders using a basket, whereas fruits such as watermelons, gherkins, cucumbers, and melons are sold per unit according to size of the fruit after harvesting (the larger, the higher the price). Leafy vegetables are sold by land portion (about 5 m^2 of area) on the farm to wholesalers. Fruit and vegetables are sold on inside and outside markets. This system of selling per basket is not kg-based and this easily allows for overlading of the units, affecting the quality of the produce and resulting in post-harvest losses (PHL).

The local markets are close to the production site. More distant markets are in other communes, specifically in larger towns such as Tokpa in Cotonou, Togba and Akassato in Abomey-Calavi. Distant markets are targeted when farmers cannot find local buyers during periods of abundance. During periods of abundance, the sale price is lower, as farmers accept the price fixed by the traders on these markets. In periods of scarcity, wholesalers visit the farmers and they define sales agreements on the spot.

2.2. Participants and sampling approach

The study targeted midstream actors, including wholesalers, retailers, transporters, and farmers active in the tomato value chain. Five wholesalers were selected as primary participants using a snowball sampling technique. These wholesalers facilitated the identification of other midstream actors in the same value chain. Consent and availability to participate in all study activities were obtained from participants, including stakeholder workshops.

2.3 Study design and implementation

The study followed a participatory and comparative design, involving three key steps (preparation phase, implementation phase and final phase).

- Preparation phase: The preparation phase focused on mobilizing stakeholders from value chains of tomato through workshops and exploratory interviews, making the reusable plastic crates and measurement tools available, and developing questionnaires to collect data on socioeconomic indicators, weight and quality of tomatoes, and perceptions of packaging methods. Enumerators were trained to ensure accurate data collection and adherence to the study's protocol. The questionnaire and study protocol were based on a previously used structure as used in other PHL measurement projects implemented by WUR (see for example Kok et al., 2019).
- Implementation phase: During the implementation phase from the mid of October to the mid of November, tomatoes were packed in reusable plastic crates, weighed baskets and

not filled above the top of the basket, and traditionally packed baskets without weighing (baseline). See picture 1A below with the different packing types. The weighing was added as an alternative solution to overcome the common practice of over-packing. Initial and final weights were measured at key points: farm level, wholesale level, and at the retail level. A specific quality grading system was developed to assess the quality of tomatoes during the pilot intervention at farm level to retail level. This was based on the quality grades applied by retailers at the final market so that we could assess the actual retail value at the different stages in the value chain (see picture 1B below with the different quality grades). During the implementation, 2 units of reusable plastic crates, 2 units of weighted baskets, and 2 units of traditional packed baskets without weighing were tracked per value chain. This was repeated twice. The trained enumerators tracked each package's journey, collecting data on weight loss, transportation costs, and economic returns at various stages in the value chain. Each package was labelled to allow identification and trace it during transportation

- Final phase: The final phase involved data analysis and reporting. Descriptive statistics were used, and postharvest losses were calculated by comparing initial and final weights per value chain. Economic feasibility and social perceptions were analyzed, and findings were shared through a final feedback workshop with all involved participants (December 2024) to capitalize on lessons learned and share the results of the pilot with the participants.

Picture 1: Different packing options (Left) and quality grades (Right) (source authors)





3. Results

3.1 Descriptive results

This section presents an overview of participant profiles based on descriptive information. In total 18 value chain actors participated of which 39% were women. The farmers and transporters were mainly men. The main markets they delivered their products to are Pahou and Kpassè markets in Ouidah.

Table 1: Participants in the study

Actor category	Number	Female % (No.)
Farmers	5	0
Wholesale traders	5	60% (3)
Retailers	5	80% (4)
Transporters	3	0
Total	18	39% (7)

The participants were linked in 5 different value chains (VC) that are described in terms of transportation time to the various markets in table 2. In Fonsa 2 farmers from different locations participated in the pilot.

Table 2: Location, time and distance to the market, tomato variety per value chain

Value chain	Location (village and market)	Transportation time, round 1 (hrs)	Transportation time, round 2 (hrs)	Average distance (kms)	Tomato variety
1	Djebadji to the market of Kpassè	14:01	12:08	4.1	Akikonkui
2	Fonsa to the market in Pahou	3:07	13:05*	11.3	Cobra
3	Fonsa to the market in Pahou	1:05	1:53	9.0	Padma/Anita
4	Degoue to the market in Kpassè	15:01	15:00	10.0	Tounvi/Akikonkui
5	Avleketé to the market in Pahou	5:30	5:50	6.6	Cobra/Akikonkui/Tounvi

*The wholesaler received first the product at her home and took it the next day to the market. She proceeded like that because she could not get enough tomatoes from the producers; so, the next day additional tomatoes were taken from other producers before taking the whole lot in the market.

Picture 2: Different packing options (Left) and means of transportation during the pilot in value chain 5 of tomatoes from Avleketé (Right) (source authors)





The baskets and crates used during the intervention had a weight ranging between 12.5 and 47.6 kg. Implying a significant variation in the quantity of produce supplied and related pressure on the tomatoes during transport. Some farmers opted for smaller-sized baskets, contributing to inconsistencies in measurement. Additionally, the traditional baskets were not always the same size and the reusable plastic crates were not consistently filled during the intervention, further impacting uniformity in the quantities measured. In addition, the limited supply of tomatoes during the second round, combined with poor weather conditions, also contributed to deviations in the measurements.

Table 3: Average weight of the different baskets and crates (kg) at farm level

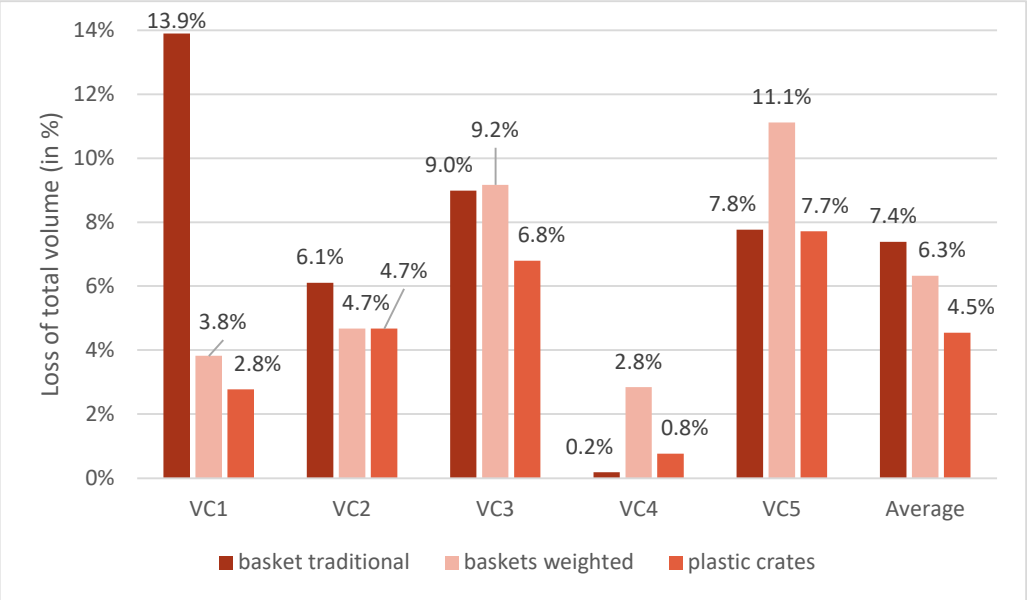
		VC 1	VC 2	VC 3	VC 4	VC 5	Average
Round 1	basket traditional (n=2)	26.2	47.7	47.6	27.9	44.7	38.8
	baskets weighted (n=2)	14.2	42.9	36.4	20.6	30.6	28.9
	plastic crates (n=2)	24.9	43.0	38.3	34.1	33.1	34.7
Round 2	basket traditional (n=2)	31.6	42.9	34.1	26.9	37.5	34.6
	baskets weighted (n=2)	12.5	34.9	28.3	19.7	27.1	24.5
	plastic crates (n=2)	24.0	42.7	38.8	27.0	38.0	34.1

3.2 Pilot results

Post-harvest losses

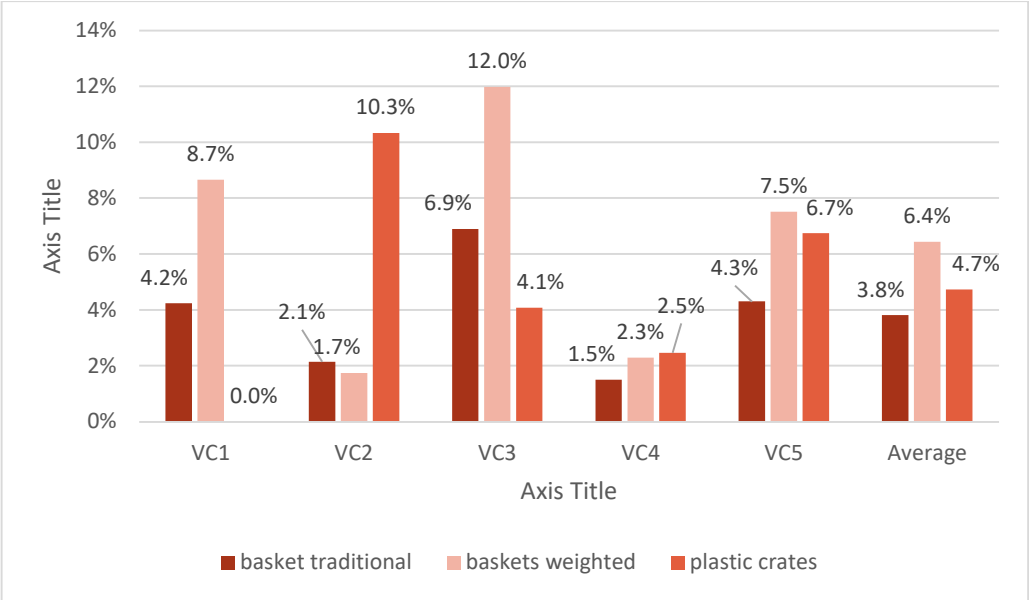
The study results show mixed outcomes, revealing limitations of the intervention due to contextual factors. The results of the first round showed variability in postharvest losses across piloted packaging methods for the different value chains, see Figure 1. While on average the reusable plastic crates demonstrated slightly lower losses compared to traditional baskets, the differences were not significant.

Figure 1: Tomatoes from farm to retailer total volume losses (%) - round 1



The results of the second round showed even greater variability in postharvest losses (PHL) across piloted packaging methods for the different value chains. On average the reusable plastic crates demonstrated even higher losses compared to traditional baskets. The second round of data collection faced also more external influences; a lack of tomatoes available due to poor weather conditions affected the tomatoes for all types of piloted packing options due to heavy rains. Farmers experienced delays before they could fill the units. This impacted the data collected and results. See the Figure below for an overview of the different observations in the second round.

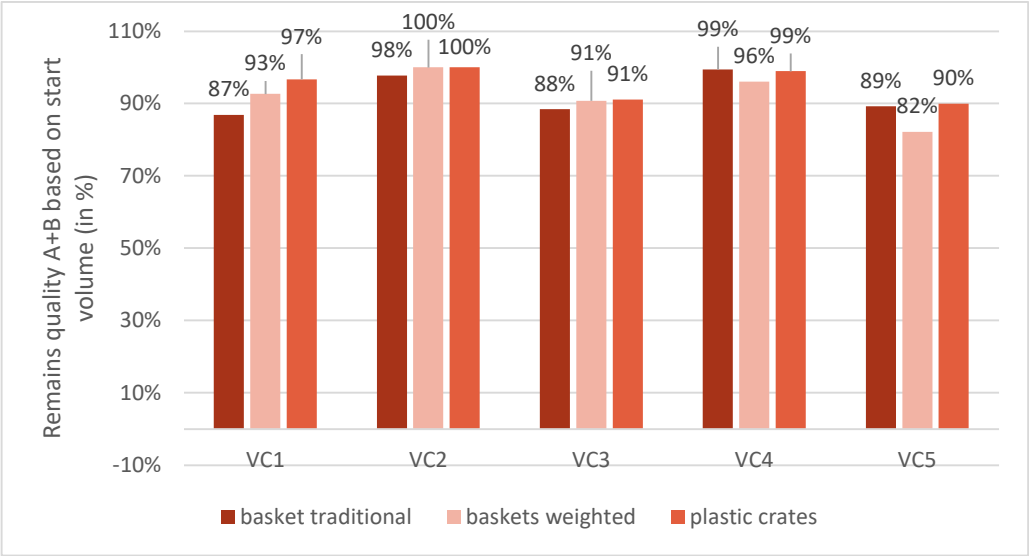
Figure 2: Tomatoes from farm to retailer total volume losses (%) - round 2



Reduced quality loss

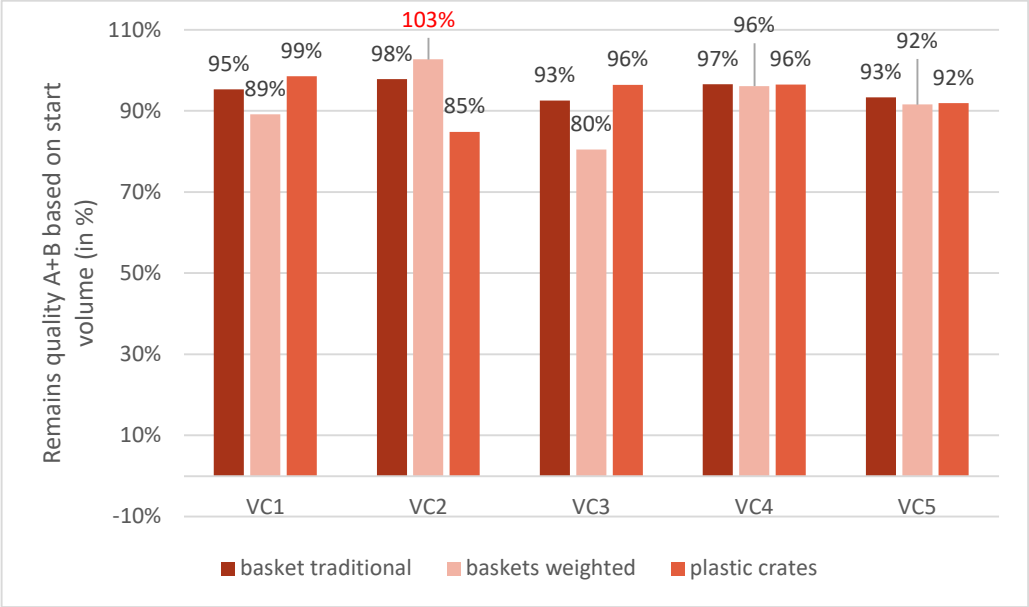
Quality losses are observed in all value chains (Figure 3). It shows the percentage performance of grade A+B tomatoes across five value chains in "Round 1" for the three different packaging types. Reusable plastic crates show the highest performance in all value chains, with an average of 95% of the quality being preserved. In 3 out of 5 value chains the weighted baskets demonstrate a higher performance than traditional baskets, averaging 92%.

Figure 3: Quality preservation (%) of A+B grade tomatoes - round 1



The second round shows mixed results due to the poor weather conditions, as described before. Overall, there is a positive average result for only VC1 and VC3 for the reusable plastic crates compared to the traditional basket packing type (see Figure 4). All other value chains show results in favor of the traditional basket (Data VC2 weighted basket is incorrect, as there is an increase in quality).

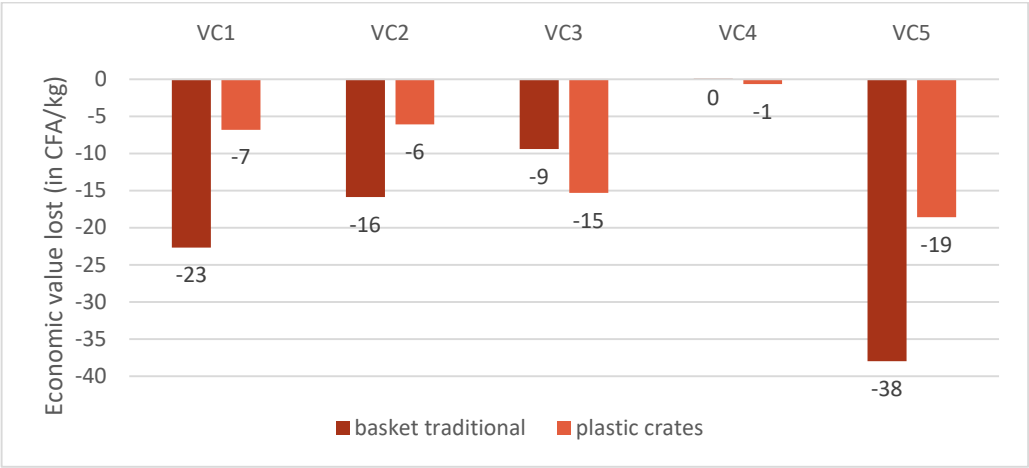
Figure 4: Quality preservation (%) of A+B grade tomatoes - round 2



Economic value assessed

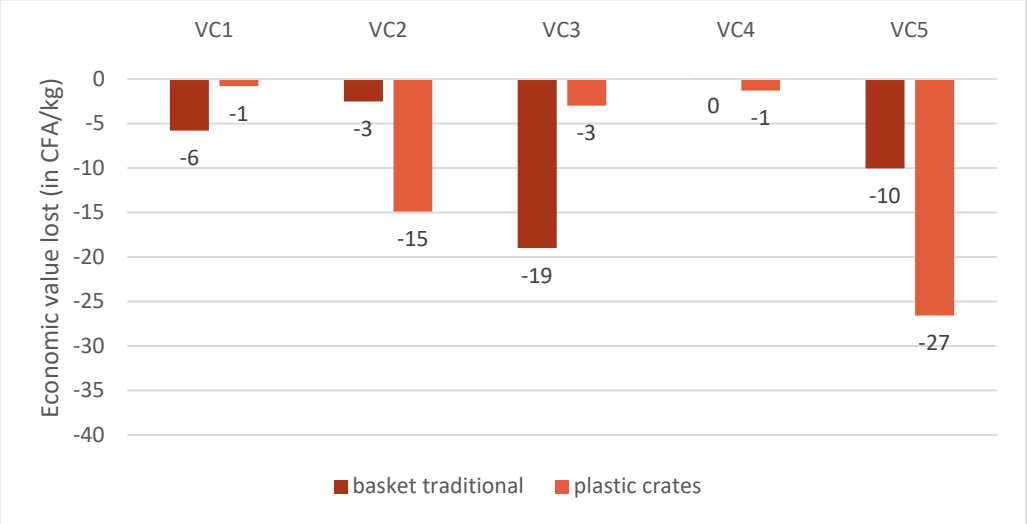
The economic losses incurred due to the proportion of tomatoes lost during transportation have been quantified (Figure 5). In the first round of assessments, four out of five value chains demonstrated higher monetary losses when using traditional baskets compared to reusable plastic crates. This indicates an opportunity to enhance value for all stakeholders in the value chain by adopting improved packaging solutions. Economic value lost

Figure 5: Economic value lost (West African Franc, CFA/kg) from farm to retailer - round 1



The second round of data collection reveals contradicting results (Figure 6). While these findings have their origin in the previous results regarding the extent of loss (measured in volume, see the previous section), they also highlight variations in the economic value lost. Interestingly, in this case, the use of reusable plastic crates, which are praised for their potential to reduce physical losses, appears to result in an average higher economic loss compared to traditional baskets. This unexpected outcome suggests that while reusable crates may mitigate some types of damage, other factors—such as supply and weather conditions—may influence their overall economic impact.

Figure 6: Economic value lost (West African Franc, CFA/kg) from farm to retailer – round 2



3.3 Findings from feedback workshop and opportunities for scaling

Adoption barriers

Stakeholder feedback revealed several logistical challenges associated with the use of reusable plastic crates. These included issues with crate availability and the difficulty of integrating their use into existing workflows. Some participants questioned the feasibility of adopting plastic crates under current market conditions, particularly given concerns about limited tomato availability during the study period. Additionally, the proximity of farms to markets reduced the potential benefits of improved packaging, as shorter transport durations minimized spoilage risks.

The study's findings were further constrained by a limited sample size and seasonal challenges. Stakeholders raised concerns about the scalability of reusable plastic crates, particularly in the absence of financial incentives such as government subsidies. They emphasized the need for further testing under varying conditions and distances to better assess the potential benefits of crate adoption. See Table 1 for an overview of the observations on suggested recommendations.

Table 4: Participants observations and recommendations

Category	Observations	Recommendations suggested
Crate availability	Limited crate availability and integration difficulties.	Ensure crates are widely available, possibly through subsidies. Encourage small-scale adoption through donations or subsidies to demonstrate benefits.
Seasonal constraints	Limited tomato availability during the study period; study conducted at the end of the harvest season	Start interventions earlier in the harvest season (July–September).
Transport conditions	Suboptimal transport led to tomato spoilage; crates not handled properly.	Train transporters to handle crates carefully, avoid overloading, and limit transport speeds.
Varietal susceptibility	Some are perceived to have higher losses.	Consider varietal differences in future assessments.
On farm damage	Heavy rot caused by ripeness, insect damage, or contamination before transportation.	Harvest tomatoes early in the morning or late in the afternoon to reduce stress and improve freshness.
Fertilizer overuse	Excessive chemical fertilizer use caused variations in tomato quality.	Educate farmers on sustainable fertilizer practices.
Packaging and stacking issues	Overfilled crates caused damage during stacking.	Avoid overfilling crates to minimize damage.
Scalability concerns	Stakeholders questioned feasibility under current market conditions; lack of financial incentives might hinder adoption.	Provide financial incentives (e.g., subsidies) and scale adoption through small-scale pilot programs. |
Crate sizes	Crate sizes do not align with traditional baskets.	Standardize crate sizes to align with traditional baskets.

Stakeholder perception of reusable plastic crates

The discussion during the workshop highlighted both the advantages and limitations of reusable plastic crate use compared to traditional baskets.

Advantages mentioned:

- Elimination of the need for wood supports during loading.
- Reduced effort required for tying plastic crates.
- Easier stacking compared to baskets.
- Protection of tomatoes from direct damage by ropes.

Limitations mentioned:

- Resistance from traders to accept equivalent quantities between crates and baskets.
- Challenges in transporting larger quantities of crates, particularly for smaller vehicles like tricycles.

Scaling up reusable plastic crate adoption

Stakeholders emphasized the importance of engaging a wide range of actors to scale up the use of reusable plastic crates (awareness raising). Key players mentioned include:

- Local extension services.
- Regional agricultural development agencies (ATDA).
- Partner projects such as SafeVeg, Support Project for the Development of Market Gardening (PADMAR), and AgriSud.
- National and local government bodies, including the ministries of agriculture and trade.
- Private investors to rent-out crates to value chain actors, as a business.
- National Federation of gardener Organizations of Benin (FeNOMA) to coordinate the standardization of unit sizes at the national level.

They also mentioned to consider establishing monitoring committees at the district level to oversee crate usage and discourage reverting to traditional baskets.

3.4 Feasibility assessment

The business model for reusable plastic crates in Benin faces significant challenges in availability, cost, and leadership for implementation. Currently, there is no local production of reusable plastic crates due to a lack of demand, which has deterred manufacturers of plastic materials from entering this market. As a result, crates must be imported from Ghana or Nigeria, driving up costs to a minimum of 6,000 CFA (or USD10) per crate, making them prohibitively expensive for value chain actors to purchase. This relatively high cost poses a major barrier for adoption, particularly in a context where access to finance is limited. Leadership in implementing this model remains unclear. However, experiences from other contexts, such as Nigeria, suggest that external investors—often from outside the tomato value chain —can play a pivotal role by introducing crate rental services, reducing the upfront costs for local actors. As highlighted in the forthcoming study by Becker et al., such rental services provide a viable pathway to overcome the financial and logistical barriers associated with direct crate ownership, offering a potential solution for the Benin context. In Nigeria, the widespread adoption of reusable plastic crates was driven by several factors: the identification of a profitable business opportunity by crate renters, the complementary benefits of reduced postharvest losses, and the immediate, positive return on investment experienced by users across the value chain. The initiative covered the North, Nigeria's tomato production hub, where significant seasonal losses necessitated innovative solutions for transporting tomatoes to southern markets (Becker et al., forthcoming).

In Nigeria, the emergence of crate renters as new entrants into the value chain was pivotal. These investors, many of whom were already active in tomato trading, recognized the growing demand for reusable plastic crates and formed an association to coordinate their activities. The association ensured exclusivity by negotiating MOUs with major plastic manufacturers to create unique crate designs and colour codes reserved for their members. This not only protected their business but also facilitated the identification and return of crates.

The association established an efficient system for returning crates from southern markets to the North. Coordinators at southern markets, well-connected with market leadership, managed crate collections and arranged transport for large volumes (minimum 1,500 crates) back to aggregation centers in the North. Crate renters recovered their investments after just seven uses per crate, with crates used up to four times per month, translating into profitability within two months. In Nigeria the crates had a lifespan of up to two years and beyond, making the business lucrative despite risks such as loss, damage, or delays.

In Benin, the lack of local manufacturing and high costs per crate make it difficult for individual value chain actors to adopt reusable plastic crates. However, replicating the Nigerian model, where investors from outside the sector lead the introduction of crate rental services, could be transformative. Establishing a crate renters' business in Benin could address operational challenges and ensure efficient collection, return, and redistribution of crates. Additionally, highlighting the reduction of postharvest losses and potential profitability could attract investors, while strategic partnerships with manufacturers or importers could ensure a steady supply of crates with unique designs.

4. Conclusion and discussion

4.1 Conclusion

The study highlights the challenges of introducing reusable plastic crates as an alternative packaging solution in South Benin's tomato value chain. The study produced mixed outcomes, making it difficult to state that reusable plastic crates perform better compared to traditional baskets. The study also revealed limitations of the intervention due to contextual factors.

- Does the use of reusable plastic crates reduce tomato post-harvest losses compared to the use of traditional baskets?

The results showed variability in post-harvest losses across packaging methods. While reusable plastic crates demonstrated slightly lower post-harvest losses compared to traditional baskets (in round 1), the differences were not significant. Loss percentages from farm to retail level were inconsistent, with in some cases no clear trend favoring crates over traditional methods. The distance to the market might have influenced this and the proximity of farms to markets reduced the potential impact of improved packaging, as shorter transport durations minimized spoilage risks. The limited sample size, the differences in packing, the difference in weight between the units and seasonal constraints also limit the robustness of the findings.

- Does the use of reusable plastic increase the value of the tomatoes traded in the value chain compared to traditional baskets?

Our study indicates a small opportunity to increase value across all stakeholders in the value chain by adopting improved packaging solutions. In the first round of assessments, four value chains demonstrated significantly higher monetary losses when using traditional baskets compared to reusable plastic crates. Data collection in the second round was affected by poor weather conditions and showed contradicting results.

While crates preserved quality slightly better, the marginal improvement in prices did not consistently offset the additional costs associated with their use. Participants noted that the initial investment in crates posed a significant barrier, especially given the limited observed advantages during the pilot.

- How are the plastic crates perceived by the stakeholders in the tomato value chain and what are the opportunities for scaling?

Feedback highlighted logistical challenges with crate availability and the need for better integration into existing workflows. Some participants had concerns about limited tomato availability during the study period and this further complicated the data collection, particularly during the second round of data collection.

- What is a feasible business model for the implementation of this innovation?

Developing a sustainable business model for the returnable plastic crate intervention poses challenges. Implementing requires substantial investments, which are beyond the financial capacity of the participants. As a result, exploring alternative funding mechanisms or business models is essential to ensure the project's feasibility and long-term success.

4.2 Discussion

The study revealed two distinct categories of issues that we address in the discussion: limitations related to the study design and challenges associated with the practical uptake of the proposed innovations.

Limitations of the study design

The small sample size was a significant limitation, restricting the generalizability of the findings. Additionally, the proximity of the farmers' markets to the farms minimized transit times, reducing the potential for fruit deterioration during transport. As a result, the outcomes were influenced by the initial quality of the produce at the farm level rather than the impact of the interventions during transport and handling.

Variability in the quantities supplied also posed a challenge, as some farmers used smaller baskets, and the crates were inconsistently filled, further complicating efforts to achieve uniformity in monitoring the effects of the various practices.

It would be valuable to repeat the pilot during both the peak supply period and over a longer distance to identify benefits. Also when tomatoes are more expensive, the potential economic benefits of packing them in reusable plastic crates are likely to be more pronounced.

Challenges regarding uptake

The study also highlighted several barriers to the practical adoption of the innovations. One critical issue was the lack of local reusable plastic crate manufacturers, which made crates prohibitively expensive for many farmers. Additionally, the size of the crates required adjustment to better align with user needs and transport requirements. Participants noted that the high initial investment needed for crates posed a significant financial barrier, particularly in the absence of clear short-term

advantages. These economic constraints underscored the need for addressing socioeconomic barriers to adoption.

To encourage broader uptake, future efforts should focus on exploring ways to make innovations like reusable plastic crates more accessible and affordable while also testing under a wider range of conditions. Socioeconomic factors, such as the cost-benefit perception among farmers and midstream actors, must be considered alongside strategies to enhance awareness, improve affordability (e.g. financial incentives from the government like subsidies) and address logistical challenges and ownership issues. These steps are essential to ensure the long-term success of post-harvest loss reduction initiatives.

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